

Table 1 Percentage Un-ionized Ammonia in Aqueous Solution by pH Value and Temperature
Calculated from data in Emerson, et. al*

pH	Temperature (°C)																4	6	8	10	12	14	16	18	20	22	24	26	28	30	32
7.0	0.11	0.13	0.16	0.18	0.22	0.25	0.29	0.34	0.39	0.46	0.52	0.60	0.69	0.80	0.91	1.01	1.11	1.21	1.31	1.41	1.51	1.61	1.71	1.81	1.91	2.01	2.11	2.21	2.31	2.41	2.51
7.2	0.18	0.21	0.25	0.29	0.34	0.40	0.46	0.54	0.62	0.70	0.78	0.86	0.96	1.06	1.16	1.26	1.36	1.46	1.56	1.66	1.76	1.86	1.96	2.06	2.16	2.26	2.36	2.46	2.56	2.66	2.76
7.4	0.29	0.34	0.40	0.46	0.54	0.63	0.73	0.85	0.98	1.14	1.31	1.50	1.73	1.98	2.26	2.56	2.86	3.16	3.46	3.76	4.06	4.36	4.66	4.96	5.26	5.56	5.86	6.16	6.46	6.76	7.06
7.6	0.45	0.53	0.63	0.73	0.86	1.00	1.16	1.34	1.55	1.79	2.06	2.36	2.71	3.10	3.53	3.96	4.36	4.76	5.16	5.56	5.96	6.36	6.76	7.16	7.56	7.96	8.36	8.76	9.16	9.56	9.96
7.8	0.72	0.84	0.99	1.16	1.35	1.57	1.82	2.11	2.44	2.81	3.22	3.70	4.23	4.82	5.48	6.14	6.79	7.43	8.07	8.71	9.35	9.99	10.63	11.27	11.91	12.55	13.19	13.83	14.47	15.11	15.75
8.0	1.13	1.33	1.56	1.82	2.12	2.47	2.86	3.30	3.81	4.38	5.02	5.74	6.54	7.43	8.42	9.50	10.58	11.66	12.74	13.82	14.90	15.98	17.06	18.14	19.22	20.30	21.38	22.46	23.54	24.62	25.70
8.2	1.79	2.10	2.45	2.86	3.32	3.85	4.45	5.14	5.90	6.76	7.72	8.80	9.98	11.29	12.72	14.14	15.56	17.00	18.42	19.84	21.26	22.68	24.10	25.52	26.94	28.36	29.78	31.20	32.62	34.04	35.46
8.4	2.80	3.28	3.83	4.45	5.17	5.97	6.88	7.90	9.04	10.31	11.71	13.26	14.95	16.78	18.77	20.80	22.82	24.84	26.86	28.88	30.90	32.92	34.94	36.96	38.98	40.99	43.01	45.03	47.05	49.07	51.09
8.6	4.37	5.10	5.93	6.88	7.95	9.14	10.48	11.97	13.61	15.41	17.37	19.50	21.78	24.22	26.80	29.42	32.04	34.66	37.28	39.90	42.52	45.14	47.76	50.38	52.99	55.61	58.23	60.85	63.47	66.09	68.71
8.8	6.75	7.85	9.09	10.48	12.04	13.76	15.66	17.73	19.98	22.41	25.00	27.74	30.62	33.62	36.72	39.82	42.92	46.02	49.12	52.22	55.32	58.42	61.52	64.62	67.72	70.82	73.92	77.02	80.12	83.22	86.32
9.0	10.30	11.90	13.68	15.65	17.82	20.18	22.73	25.46	28.36	31.40	34.56	37.83	41.16	44.53	47.91	51.28	54.65	58.02	61.39	64.76	68.13	71.50	74.87	78.24	81.61	84.98	88.35	91.72	95.09	98.46	101.83
9.2	15.39	17.63	20.08	22.73	25.58	28.61	31.80	35.12	38.55	42.04	45.57	49.09	52.58	55.99	59.31	62.63	65.95	69.27	72.59	75.91	79.23	82.55	85.87	89.19	92.51	95.83	99.15	102.47	105.79	109.11	112.43
9.4	22.38	25.33	28.47	31.80	35.26	38.84	42.49	46.18	49.85	53.48	57.02	60.45	63.73	66.85	69.79	72.62	75.45	78.27	81.09	83.91	86.73	89.55	92.37	95.19	98.01	100.83	103.65	106.47	109.29	112.11	114.93
9.6	31.36	34.96	38.38	42.49	46.33	50.16	53.94	57.62	61.17	64.56	67.77	70.78	73.58	76.17	78.55	80.83	83.11	85.39	87.67	89.95	92.23	94.51	96.79	99.07	101.35	103.63	105.91	108.19	110.47	112.75	115.03
9.8	42.00	46.00	50.00	53.94	57.78	61.47	64.99	68.31	71.40	74.28	76.92	79.33	81.53	83.51	85.30	86.99	88.68	90.37	92.06	93.75	95.44	97.13	98.82	100.51	102.20	103.89	105.58	107.27	108.96	110.65	112.34
10.0	53.44	57.45	61.31	64.98	68.44	71.66	74.63	77.35	79.83	82.07	84.08	85.88	87.49	88.92	90.19	91.46	92.73	94.00	95.27	96.54	97.81	99.08	100.35	101.62	102.89	104.16	105.43	106.70	107.97	109.24	110.51
10.2	64.53	68.15	71.52	74.63	77.46	80.03	82.34	84.41	86.25	87.88	89.33	90.60	91.73	92.71	93.58	94.45	95.32	96.19	97.06	97.93	98.80	99.67	100.54	101.41	102.28	103.15	104.02	104.89	105.76	106.63	107.50

* Emerson, K., R. C. Russo, R.E. Lund, and R.V. Thurston. 1975. Aqueous ammonia equilibrium calculations: effect of pH and temperature. *J. Fish. Res. Board Can.*, 32:2379-2383.